VCL2FMX

APPLICATION INVENTORY

VCL2FMXConverter

v6.0 Navigator

Build 021 — Codebase, Coverage, and Release Profile

A measured inventory of source architecture, mapping intelligence, validation coverage, documentation, and release artifacts.

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Apache License 2.0 • Release commit db1d31b9

EXECUTIVE PROFILE

Build 021 at a glance

Navigator is a contract-driven Delphi conversion application with a designer-owned FireMonkey interface, extensible mapping packs, deterministic reporting, and layered validation.

33,321	493	227	62	64
IMPLEMENTATION LINES	MAPPING RULES	CONTRACTS PASSED	ACCEPTANCE CHECKS	REGRESSION GUARDS

Key inventory

Area	Build 021 measure	What it represents
Production source	24 Pascal units + 1 DPR	31,626 physical Pascal lines
Designer UI	1 MainForm.fmx	1,695 lines of IDE-editable form definition
Mapping intelligence	22 packs / 22 vendors	493 rules with property and event mappings
Contract validation	227 of 227 passed	19 categories; zero failures
Release validation	Win32 + Win64	Both Release builds succeeded
Report output	3 formats	Plain text, HTML, and deterministic JSON
Documentation tree	311 files	Guides, PDFs, notes, help, and supporting material

RELEASE POSITION

Build 021 is the validated Navigator baseline: maintainability, report integrity, designer-owned UI, mapping-pack discipline, and stronger validation are the release priorities.

PROGRAM INVENTORY

Entry points and project fixtures

The repository contains one production application and a deliberate set of developer, sample, regression, and contract project entry points.

Program category	Count	Purpose
Production GUI converter	1	VCL2FMXConverter.dpr
Developer console harness	1	RunConversionEngine.dpr
Sample projects	6	Demonstration and usage examples
Regression-test projects	2	Focused executable regression coverage
Contract-project fixtures	12	Project-level conversion behavior
Total DPR entry points	22	Excludes generated output

Production architecture

Unit group	Units	Primary responsibility
Advanced	4	Higher-order analysis and semantic groundwork
Core	4	Engine orchestration, models, options, and shared behavior
Generator	1	FMX source and form generation
Mapper	2	Component and mapping-pack decisions
Parser	2	Pascal and form input parsing
Project	1	Project-level discovery and conversion
Reporting	4	Text, HTML, JSON, and fidelity reporting
Rewrite	5	Targeted Pascal transformation passes
Application UI	1	Designer-owned MainForm implementation

SOURCE PROFILE

Codebase size and concentration

Source measure	Count
Production Pascal units	24
Production Pascal lines	31,592
Main project DPR lines	34
Production Pascal total	31,626
Designer-owned MainForm.fmx lines	1,695
Implementation footprint	33,321

Largest production units

Unit	Physical lines
Converter.Rewrite.LiveBindings.pas	3,198
Converter.Core.Engine.pas	2,650
Converter.Generator.FMX.pas	2,504
MainForm.pas	2,425
Converter.Mapper.Component.pas	2,404

INTERPRETATION

The largest units identify the best future refactoring targets. Navigator treats compiler-style architecture as a measured research track, not a v6.0 release promise.

MAPPING INTELLIGENCE

Mapping-pack coverage

Third-party knowledge is kept in validated mapping packs instead of being scattered through converter logic, preserving a disciplined and extensible conversion model.

22	493	674	494
MAPPING PACKS	RULES	PROPERTY MAPS	EVENT MAPS

Rule action distribution

Action	Rules	Share	Meaning
Convert	21	4.3%	Direct supported conversion
Partial	105	21.3%	Useful conversion with review
Manual review	100	20.3%	Human decision required
Detect only	186	37.7%	Identify and report without unsafe rewriting
Preserve	81	16.4%	Retain source behavior or declaration
Total	493	100.0%	All validated mapping rules

PACK PROFILE

The 22 vendor packs contain 4,974 JSON lines and conform to the vcl2fmx.mapping-pack.v1 schema.

CONTRACT VALIDATION

227 executable conversion contracts

Every contract passed. The suite spans 19 conversion categories and includes multi-file fixtures, expected outputs, report expectations, uses-clause behavior, and forbidden patterns.

227	0	189	38
PASSED	FAILED	COMPILER ELIGIBLE	INTENTIONAL EXCLUSIONS

Coverage by category

Category	Count	Category	Count
Colors	16	Comments and boundaries	23
Components and events	26	Data-aware	8
DFM fidelity	4	DFM pairs	10
Dry-run preview	2	Encoding	7
Graphics	8	Include analysis	11
LiveBindings	6	Pascal structure	15
Project files	6	Project integration	6
Reporting	4	Semantic resolution	4
Third-party	6	Uses clause	25
WinAPI messages	40		

QUALITY AND DELIVERY

Layered validation

Validation layer	Result	Evidence
Contract suite	227 pass / 0 fail	19 conversion categories
Acceptance suite	62 pass / 0 fail	Release-level behavior checks
Regression guards	64	Structural and policy protections
Win32 Release	Succeeded	Validated production executable
Win64 Release	Succeeded	Validated production executable

Reporting and project support

Area	Inventory
Report formats	Plain text, HTML, deterministic vcl2fmx.report.v1 JSON
Maintenance tooling	44 PowerShell scripts + 2 Python scripts; 7,672 lines including harness
Documentation	311 files: 21 DOCX, 24 PDF, 12 Markdown, 30 Help HTML pages
Local website	2 HTML pages; approximately 750 lines across pages, CSS, and JavaScript
Tracked repository	2,395 files; 1,865 tracked text files
Tracked text volume	Approximately 211,981 lines including docs, goldens, and history

REPORTING INTEGRITY

Navigator reports the distinction between conversion, partial conversion, detection, preservation, and manual review instead of overstating automatic conversion coverage.

RELEASE ARTIFACTS

Build 021 deliverables

Artifact	Size	SHA-256
Win32 Release EXE	13.36 MB	A85AB12AD41452EC5D3085095D8C131729889B684B5ECE579EC8EF4AC3FB9335
Win64 Release EXE	18.89 MB	9D5847D2AEE7030EB92BE7E23B979B8732E5C1E87270064E385D58BE071DE741
Source distribution ZIP	852,970 bytes	BABB8C2BF8F1EC1DD6DB9F3784D0509B09FCB5603B37F677D5C4738D82E21FBF

Source distribution

VCL2FMXConverter_v6_0_Navigator_Build021_Source_Distribution_2026-08-02_133819.zip

Measurement notes

Measure	Method
Line counts	Physical lines, including comments and blank lines.
Implementation footprint	Production Pascal plus the designer-owned FMX form definition.
Exclusions	Generated conversion output and build-output folders are excluded from program counts.
Repository estimate	Tracked text volume includes documentation, contract goldens, and historical material; it is not application code size.
Validation record	The latest field-trial record is docs/notes/V6_BUILD021_FIELD_TRIAL_RESULTS.md.

BUILD IDENTITY

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